

Multilateral Tariff Diversion in United States Machinery and Electronics Imports: A Continuous Difference-in-Differences Approach

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Replication Archive: <https://github.com/pagar-ani/us-tariff-diversion-replication>

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Abstract

Evaluating 36 consecutive months of administrative customs declarations across machinery and electronics (Harmonized System Chapters 84 and 85, $N = 25,624$ across 133 four-digit HS headings), this paper investigates global supply chain reallocation following 2025 United States reciprocal border duties. We examine whether discriminatory tariffs induced domestic reshoring or multilateral trade diversion toward alternative production hubs. Under Chapter 99 provisions, advanced computing and telecommunications apparatus entering from non-China partners were exempted from reciprocal tariffs, while Chinese shipments faced cumulative duties exceeding 60%. Implementing a continuous difference-in-differences design using an exogenous pre-panel baseline (ChinaShare²⁰²²), we estimate a multilateral diversion elasticity relative to China of $\hat{\beta}_{\text{ExChina}} = 1.4720$ ($SE = 0.2886$, $p < 0.0001$). Regression discontinuity in time splines and dynamic event studies establish pre-treatment stationarity followed by a discrete enforcement step jump ($\hat{\delta} = 1.3673$, $p < 0.0001$). Sourcing reallocation is pervasive across capital goods outside computing and telecommunications apparatus ($\hat{\beta}_{\text{NonTech}} = 1.2095$, $p = 0.0001$). We detect no statistically significant change in landed unit values along the exposure gradient ($\hat{\beta}_{\text{P}} = +0.1181$, $p = 0.1631$), consistent with proportional volume expansion across established overseas assembly hubs.

JEL Codes: F13, F14, C23, L63.

Keywords: Reciprocal Tariffs, Multilateral Diversion, Continuous DiD, PPML, Exogenous Baseline, Global Value Chains.

1. Introduction

On April 2, 2025, United States Executive Order 14257 (United States Executive Office of the President, 2025a) enacted reciprocal border duties citing persistent goods trade deficits (90 FR 15041). While Annex I of the initial order scheduled country-specific reciprocal rates across 57 trading partners (e.g., Vietnam at 46%, alongside a 10% baseline for partners not listed in Annex I; Nguyen, Nguyen, and Boudreau, 2025), subsequent executive actions restructured these border barriers. On April 9, 2025, Executive Order 14266 (United States Executive Office of the President, 2025c) established a 90-day suspension of enhanced Annex I rates for non-China partners—temporarily setting baseline reciprocal duties at 10%—while explicitly excluding the People's Republic of China from relief. Under Annex II of Executive Order 14257 (codified under HTSUS Chapter 99 subchapter III, secondary code 9903.01.32) and the Presidential Memorandum of April 11, 2025 ("*Clarification of Exceptions*"; United States Executive Office of the President, 2025d), advanced information technology hardware was explicitly exempted from reciprocal duties altogether. Under the WTO Information Technology Agreement (ITA-1, 1996), statutory Most-Favored-Nation (MFN) duties on computing machinery (Heading 8471) and telecommunications apparatus (Heading 8517) have stood at 0.0% in the United States since 1997. Because Annex II exempted these classifications from reciprocal surcharges, and Mexican shipments entered duty-free under USMCA rules of origin, electronic hardware entering from non-China trading partners faced an effective statutory tariff of $\tau(i, t) = 0.0\%$.

In contrast, identical electronic hardware originating from the People's Republic of China was excluded from exemption relief and remained encumbered by cumulative Section 301 List 1–4B duties and national security revisions exceeding 60% ($\tau(\text{CHN}, t) \geq 60.0\%$). Consequently, United States purchasing managers faced an asymmetric relative price wedge:

$$\Delta\tau_{it} = \tau(\text{CHN}, t) - \tau(i, t) \approx 60.0\% - 0.0\% = +\mathbf{60.0\%} \quad (1)$$

During the April–July 2025 window, non-exempt general machinery from alternative partners faced the temporary 10% reciprocal rate, while Chinese shipments faced over 60%, maintaining a net relative tariff wedge on China of +50.0% or greater. Following executive rate modifications in August 2025 (Executive Order 14326; United States Executive Office of the President, 2025b) and subsequent procedural scoping revisions under Executive Order 14346 (United States Executive Office of the President, 2025e), partner-specific schedules ranged from 19% to 50%, sustaining substantial relative wedges against Chinese supply. Quantitative trade general equilibrium models of the 2025 reciprocal tariff shock (Rotunno and Ruta, 2025) establish that bilateral penalties of this magnitude induce severe trade contraction on the affected origin (–70% to –88%) and deflect trade toward third-country markets. Our continuous difference-in-differences empirical strategy does not impose a structural assumption on absolute foreign tariff levels; identification operates through the relative price penalty ($\Delta\tau_{it} \gg 0$) interacted with pre-determined sectoral exposure.

Relational supply chain search-and-bargaining frameworks (Grossman and Helpman, 2021) predict that when a buyer faces discriminatory tariffs across competing supply footprints, ongoing relational contracts undergo renegotiation and purchasing volumes reallocate toward zero-duty foreign suppliers (Vinerian trade diversion, whereby trade shifts from a lower-cost duty-encumbered origin to higher-cost or alternative duty-free suppliers). In high-technology electronics, global electronic manufacturing services (EMS) providers (such as Foxconn, Quanta, Pegatron, Wistron, Hosiden, and Goertek) already maintained operational campuses across East Asia and North America. Rather than repatriating capital-intensive fabrication to the United States, corporate procurement divisions shifted volume quotas toward overseas nodes. Customs data and press reporting suggest this reallocation in consumer hardware: following cumulative duties exceeding 60% on Chinese assembly, China's share of United States video game console imports fell from 86% in 2024 to under 5% by mid-2025, while Vietnamese shipments expanded from less than \$30 million to over \$400 million per month as contract manufacturers expanded assembly facilities in Bac Ninh and Hai Phong (Bloomberg News, 2025). This rapid reshuffling among established manufacturing partners corroborates the macro-level reallocation patterns documented by Alfaro and Chor (2025), who show that United States decoupling from China

accelerated into an intensive-margin reallocation among top-20 import nodes—reducing China's share of United States sourcing to 9% by July 2025.

Our empirical estimator identifies the *differential* reallocation of trade across sectors with varying baseline Chinese exposure relative to China, holding common global demand and country-specific macro shocks constant. We distinguish this econometric identification from gross dollar accounting: while non-China imports expanded by +\$24.64 billion per month post-tariff, aggregate trade flows also absorbed secular United States macroeconomic demand, exchange rate shifts, and the high-tech AI investment cycle (International Monetary Fund, 2025a). Open-economy production network models (Kalemli-Özcan, Soylu, and Yildirim, 2025) demonstrate that reciprocal tariffs propagate through global input-output linkages to generate cross-border production distortions. By utilizing sector-by-period and partner-by-period fixed effects, our continuous difference-in-differences estimator isolates the specific tariff-induced relative reallocation component from these broader macroeconomic drivers.

Concurrent export controls promulgated by the Department of Commerce Bureau of Industry and Security in October 2022 and October 2023 (United States Department of Commerce, 2023) restricted outbound shipments of advanced semiconductors and fabrication equipment to China rather than United States import arrivals (Ando, Hayakawa, and Kimura, 2024); the stationarity of our 2023–2024 pre-treatment baseline (Appendix Table A1) provides no evidence of confounding pre-trends prior to tariff enactment.

Using 36 consecutive months of administrative customs declarations from the United Nations Comtrade Database (United Nations Statistics Division, 2026, N = 25,624), this paper investigates how reciprocal border duties restructured trade flows across foreign assembly hubs, and demonstrates how comparative RDiT splines and multi-estimator frameworks isolate policy-induced reallocation across global value chains.

2. Sample Construction

We study all 133 active four-digit classifications in Harmonized System Chapters 84 and 85 (87 headings in Chapter 84 and 46 headings in Chapter 85) that exhibited positive United States import value during the 2022 baseline calendar year. Our focus on Harmonized System Chapters 84 and 85 is governed by institutional policy architecture and measurement consistency: (i) Chapter 99 subchapter III exemptions (under Headings 8471 and 8517) were explicitly tailored to electronic computing and telecommunications apparatus; (ii) continuous exposure (ChinaShare_{s,2022}) is constructed across the complete universe of 133 advanced machinery and electronics headings, ensuring common support within high-technology manufacturing; and (iii) non-parametric exposure terciles exploit within-universe variation across capital-goods sectors facing identical supply-chain frictions. Pooling all 97 HS chapters would dilute institutional identification, introduce severe unobserved non-tariff border measures, and confound capital-goods value chain reallocation with unrelated commodity-specific tariff regimes.

The panel covers bilateral customs declarations across 36 consecutive calendar months from January 2023 through December 2025 (T = 36), partitioned into a 27-month pre-tariff window (2023M01–2025M03) and a 9-month post-tariff window (2025M04–2025M12). We evaluate declarations for the six primary machinery and electronics exporters to the United States: China (the affected origin subject to duties), alongside the five primary alternative sourcing partners: Taiwan, Mexico, Vietnam, Thailand, and India. From an estimand perspective, $\hat{\beta}_{\text{ExChina}}$ measures the average differential sourcing reallocation of these five primary alternative hubs relative to China along the exposure gradient. In Section 4.9, we systematically evaluate sensitivity to the partner reference group by extending estimation to an 11-country panel extract including Malaysia, Germany, Japan, South Korea, and the Philippines.

For each of the 133 sectors, baseline exposure to Chinese supply is constructed deterministically against official World imports during the **2022 calendar year**—two full years prior to the 2025 reciprocal tariff actions and one year prior to the panel start date:

$$\text{ChinaShare}_s^{2022} = \frac{\sum_{t \in 2022} \text{ImportValue}(\text{CHN}, s, t)}{\sum_{t \in 2022} \text{ImportValue}(\text{World}, s, t)} \quad (2)$$

where World denotes total United States import declarations across all foreign supplier origins in sector s during 2022, ensuring the exposure denominator represents China's pre-tariff market share within the United States. Baseline shares range continuously from 0.14% (HS 8401) to 95.65% (HS 8510), with a cross-sectoral mean of 19.59% (SD = 16.96%). The structural correlation between 2022 and 2024 China shares is $\rho = 0.949$, consistent with persistent comparative advantage; because exposure is measured in 2022, it pre-dates the 2025 policy actions. The resulting panel comprises $133 \times 6 \times 36 = 28,728$ cells, of which 25,624 exhibit positive customs transactions and 3,104 represent zero-trade cells. Customs declarations reflect the official United Nations Comtrade database (United Nations Statistics Division, 2026; Reporter Code: 842 USA), extracted February 2026. While administrative customs data undergo routine reconciliation across agency reporting vintages, historical revisions for advanced manufactured goods in Chapters 84 and 85 are bounded within minor tolerances ($\pm 0.5\%$).

Regarding the legal mechanics of tariff exemptions, United States Customs and Border Protection (CBP) Cargo Systems Messaging Service (CSMS) guidance and executive implementing memoranda following Executive Order 14257 (specifically CSMS #64649265 of April 4, 2025, CSMS #64680374 of April 8, 2025, and CSMS #64724565 of April 11, 2025; United States Customs and Border Protection, 2025) established that Chapter 99 exemptions for advanced machinery and telecommunications hardware entering from non-China origins were strictly classification-based. Under general customs law (19 CFR § 134.1), country-of-origin determinations follow the substantial transformation doctrine rather than statutory Regional Value Content (RVC) percentage thresholds. Consequently, foreign EMS providers operating in Vietnam, Mexico, Taiwan, Thailand, and India were legally authorized to import intermediate sub-assemblies originating in China, integrate them into finished capital goods (undergoing a customs tariff shift, e.g., from intermediate parts under Heading 8517.70 to finished equipment under Heading 8517.13), and enter the United States exempt at $\tau = 0\%$. This created an institutional divergence between customs law and trade diplomacy: while third-country assembly legally complied with substantial transformation, United States trade negotiators aggressively challenged third-country reliance on Chinese intermediate components—threatening enhanced 40% transshipment duties on goods routed through Southeast Asia codified under Section 3(a) of Executive Order 14326 (United States Executive Office of the President, 2025b; Stevens and Nguyen, 2025; Guarascio and Nguyen, 2025). In response, Vietnamese trade authorities drafted stricter origin tracing guidelines to defend legitimate manufacturing designations (Hoang, 2025; Guarascio, 2025), even as Chinese intermediate inputs accounted for an estimated 19% of Vietnamese exports to the United States (Hoang, 2025). Granular enterprise-level customs decompositions verify this structural reality: Iyoha et al. (2025) demonstrate that only 8.8% of Vietnam's export surge reflected circumvention rerouting, while 39.8% represented domestic value added and 51.4% imported intermediate inputs (20.4% from China and 31.0% from the rest of the world). Similarly, multi-region input-output synthetic control estimates establish that Vietnamese electronics exports gained 10 percentage points in domestic value-added share while Chinese direct value-added content declined (Schulze and Xin, 2025). This institutional and empirical reality accounts for why physical shipping density held strictly constant ($\hat{\beta}_{\text{density}} = -0.1181, p = 0.1768$) while third-country customs values surged.

We also evaluate whether constructing baseline exposure in 2022 introduces endogeneity bias from historical Section 301 tariffs (2018–2020). Methodologically, identification in shift-share continuous difference-in-differences requires pre-determination with respect to the specific policy shock under evaluation (Goldsmith-Pinkham, Sorkin, and Swift, 2020; Borusyak, Hull, and Jaravel, 2022). Calendar year 2022 strictly satisfies this condition, preceding the 2025 reciprocal tariff actions by two full years and pre-dating our estimation panel. For historical Section 301 diversion to bias our 2025 estimator, 2018–2022 sourcing adjustments would have to correlate with differential post-2024 trends. However, our unconstrained 24-month pre-trend tests across 2023 and 2024 display strictly

zero velocity ($\hat{\theta}_{24m} = -0.0048$, $p = 0.7397$; annual window equality $F(1, 132) = 0.1098$, $p = 0.7409$; quarterly leads $F(7, 132) = 0.8911$, $p = 0.5156$). If historical Section 301 diversion had created persistent trend momentum along the exposure gradient, it would have manifested as upward drift during the 2023–2024 baseline. The empirical stationarity of the pre-panel baseline decisively falsifies this bias channel. Moreover, cross-sector comparative advantage rankings are structurally stable across years ($\rho = 0.949$ between 2022 and 2024 shares; earlier pre-2018 baselines require concordancing across the WCO HS 2022 Nomenclature revision, such as headings 8485, 8549, and 8535/8536 splits, without altering identifying variation).

3. Multilateral Sourcing Reallocation

Table 1 decomposes monthly trade values across the pre- and post-tariff windows for all 133 four-digit HS headings in Chapters 84 and 85. These figures represent unadjusted descriptive gross customs arrivals that reflect both policy-induced reallocation and secular macroeconomic and investment trends; our econometric specifications in Section 4 isolate the relative diversion parameter.

Table 1: Descriptive Multilateral Sourcing Reallocation Across 133 HS Headings

EXPORTER / PARTNER	PRE-TARIFF (\$M/MO)	POST-TARIFF (\$M/MO)	ABSOLUTE DIFF (\$M/MO)	GROWTH (%)
<i>China (Subject to Duties):</i>				
China	\$17,541.69	\$9,714.92	-\$7,826.77	-44.62%
<i>Alternative Partners:</i>				
Taiwan	\$6,647.52	\$16,637.92	+\$9,990.40	+150.29%
Mexico	\$15,361.32	\$20,858.55	+\$5,497.23	+35.79%
Vietnam	\$5,699.02	\$10,643.24	+\$4,944.22	+86.76%
Thailand	\$3,100.72	\$5,886.36	+\$2,785.64	+89.84%
India	\$1,823.50	\$3,241.88	+\$1,418.38	+77.78%
Combined Alternative Hubs Total	\$32,632.08	\$57,267.95	+\$24,635.87	+75.50%

Source: UN Comtrade Database (Reporter Code 842 USA); author's calculations. Pre-period: 2023M01–2025M03 (27 months). Post-period: 2025M04–2025M12 (9 months). Universe includes all 133 active four-digit headings under HS Chapters 84 and 85.

Note: Alternative hubs captured \$24.6B in monthly trade diversion, more than offsetting the \$7.8B decline from China. Five alternative hubs absorbed 3.2x the volume lost from China, confirming multilateral EMS reallocation over reshoring.

The empirical patterns indicate broad-based multilateral reallocation rather than a Vietnam-specific expansion. United States imports from China contracted by \$7,826.77 million per month (-44.6%), while combined imports from the five alternative partners expanded by \$24,635.87 million per month (+75.5%). Sourcing gains were distributed across multiple hubs with established manufacturing ecosystems (Fajgelbaum et al., 2024): Taiwan captured the largest absolute gain (+\$9,990.40 million/month, +150.3%) driven by computing hardware, Mexico expanded substantially (+\$5,497.23 million/month, +35.8%), and Vietnam accounted for approximately one-fifth (20.1%) of the total non-China expansion (+\$4,944.22 million/month, +86.8%; Choi and Nguyen, 2023; Mayr-Dorn et al., 2023). Thailand and India also experienced rapid percentage growth (+89.8% and +77.8%, respectively).

Because administrative customs declarations record both total shipment value and net physical shipping mass, we evaluate landed price and shipping density margins directly (Zheng et al., 2023). Defining landed unit values as $UV_{ist} = \text{Value}_{ist} / \text{Mass}_{ist}$ (in USD/kg) and physical shipping density as $\text{Density}_{ist} = \text{Mass}_{ist} / \text{Value}_{ist}$, these two measures represent a single underlying regression in two reciprocal parameterizations:

$$\ln(UV_{ist}) \equiv -\ln(\text{Density}_{ist}) \Rightarrow \hat{\beta}_p \equiv -\hat{\beta}_{\text{density}} \quad (3)$$

Estimating our saturated three-way fixed effects specification yields no statistically significant change in landed unit values along the exposure gradient ($\hat{\beta}_{_P} = +0.1181$, $SE = 0.0842$, $t = 1.40$, $p = 0.1631$; two-way clustered $SE = 0.0752$, $t = 1.57$, $p = 0.1768$ under $t(5)$), with a similarly small, statistically insignificant response when interacting treatment with focal telecommunications apparatus (HS 8517, $\hat{\theta}_{_P} = +0.0765$, $SE = 0.0803$, $t = 0.95$, $p = 0.3845$ under $t(5)$). The estimated landed price response is statistically indistinguishable from zero, consistent with proportional volume expansion along the exposure gradient and providing no evidence of pure invoice relabeling (Amiti, Redding, and Weinstein, 2019; Fajgelbaum et al., 2020). Partner economies expanded volume at essentially unchanged landed border prices without charging an exporter markup along the exposure gradient. Enterprise disclosures and plant reporting suggest modular supply chain adjustments across established overseas assembly facilities (Bloomberg News, 2025), consistent with evidence that tariffs chilled Chinese outward greenfield FDI and encouraged retrenchment (Zeng and Kim, 2024).

4. Empirical Framework and Multi-Estimator Robustness

To evaluate whether cross-sector exposure to Chinese market share drove alternative partner import expansion, we formulate the high-dimensional multilateral difference-in-differences specification:

$$\ln(Y_{ist}) = \alpha_{is} + \lambda_{st} + \gamma_{it} + \beta_{\text{ExChina}}(\text{ExChina}_i \times \text{ChinaShare}_s^{2022} \times \text{Post}_t) + \varepsilon_{ist} \quad (4)$$

where:

- Y_{ist} denotes United States import value (in millions USD) from partner country i in four-digit HS heading s during calendar month t . For Poisson pseudo-maximum-likelihood (PPML) estimation, Y_{ist} enters in levels, incorporating zero-trade cells.
- $\text{ExChina}_i = 1 - 1\{i = \text{CHN}\}$ is an indicator equal to 1 for the five alternative sourcing hubs (TWN, MEX, VNM, THA, IND) and 0 for China (CHN). Because the sector-period fixed effects λ_{st} absorb the common term $\text{ChinaShare}_s^{2022} \times \text{Post}_t$, the regressor collapses algebraically to $-1\{i = \text{CHN}\} \times \text{ChinaShare}_s^{2022} \times \text{Post}_t$. Consequently, $\hat{\beta}_{_ \text{ExChina}} \equiv -\hat{\beta}_{_ \text{CHN}}$ identifies the differential post-tariff diversion of the non-China partner cohort relative to China along the pre-determined exposure gradient.
- $\text{ChinaShare}_s^{2022}$ is China's pre-determined share of total United States World imports in sector s during the 2022 calendar year ($\sum(t \in 2022) \text{CHN}_{st} / \sum(t \in 2022) \text{World}_{st}$), strictly pre-dating the 2023–2025 panel window.
- Post_t is an indicator equal to 1 for post-tariff months ($t \geq 2025\text{Mo4}$, April through December 2025) and 0 for pre-tariff baseline months ($t \leq 2025\text{Mo3}$, January 2023 through March 2025).
- α_{is} represents 798 partner $\times$ sector fixed effects (784 non-redundant degrees of freedom), absorbing time-invariant bilateral comparative advantage and trade costs.
- λ_{st} represents 4,788 sector $\times$ period fixed effects (4,696 non-redundant degrees of freedom), non-parametrically absorbing sector-by-period demand shifts, commodity price shocks, and common technology cycles across the observed partner matrix.
- γ_{it} absorbs 216 partner $\times$ period fixed effects (215 non-redundant degrees of freedom), controlling for partner-wide exchange rate fluctuations, macro shocks, and general shipping inflation.
- Standard errors are clustered at the four-digit sector level ($G = 133$), where continuous policy exposure varies across product headings (Abadie et al., 2023); two-way clustering by partner and sector ($G_1 = 6$, $G_2 = 133$) evaluated against conservative finite-cluster $t(5)$ critical values confirms statistical significance across all positive trade specifications following Cameron, Gelbach, and Miller (2011), MacKinnon and Webb (2017, 2018), Conley and Taber (2011), and Ferman and Pinto (2019).

We evaluate identification credibility across the multi-estimator suite reported in Table 2:

Table 2: Continuous Difference-in-Differences Robustness Suite

MODEL SPECIFICATION	ESTIMATOR	β (EX-CHINA)	SE (G=133)	SE (2-WAY)	T-STAT	P-VALUE	N
(1) OLS Baseline (2022 Share)	Standard OLS	1.4720***	0.2886	0.2862	5.10	<0.0001	25,544
(2) PPML (including zeros)	Poisson PML	3.2629***	1.1738	0.8850	2.78	0.0054	28,353
(3) RDIT Enforcement Step Jump	Comparative RDIT	1.3673***	0.2980	0.2562	4.59	<0.0001	25,544
(4) Non-Tech PPML (excl. 8471/8517)	Poisson PML	1.2095***	0.3115	0.3187	3.88	0.0001	27,921
(5) Landed Unit Values (ln UV)	Standard OLS	0.1181	0.0842	0.0752	1.40	0.1631	22,764
(6) Pre-Q1 Excl. (excl. 2025Q1)	Standard OLS	1.5416***	0.3009	0.2976	5.12	<0.0001	23,404

Source: OLS and PPML regressions on 25,624 customs declarations; author's calculations. All models include Partner $\times$ Sector, Sector $\times$ Period, and Partner $\times$ Period fixed effects. In Cols 1, 2, 4, 5, and 6, treatment is ExChina_{it} $\times$ ChinaShare_{it}²⁰²² $\times$ Post_{it}.

Note: Primary standard errors clustered at the four-digit sector level (G = 133 in Cols 1–3, 5, 6; G = 131 in Col 4). Two-way clustered standard errors by Partner and Sector (G1 = 6, G2 = 133) follow Cameron, Gelbach, and Miller (2011), MacKinnon and Webb (2017, 2018), Conley and Taber (2011), and Ferman and Pinto (2019) with conservative t(5) degrees of freedom; under t(5), all positive trade models remain statistically significant (Col 1 p = 0.0036, Col 2 p = 0.0142, Col 3 p = 0.0031, Col 4 p = 0.0128, Col 6 p = 0.0035; in Col 5, two-way t = 1.57, p = 0.1768). Wild cluster bootstrap with Webb (2014) 6-point weights (G = 133) yields p_WCB $\leq$ 0.0007. Overall R² for baseline OLS is 0.9519; within R² = 0.0045 reflects the absorption of 5,695 fixed effect parameters. *** p < 0.01.

Sample Accounting: Col 1 drops zeros and 80 singletons (N = 25,544); Col 2 incorporates zeros, dropping 375 separated cells during absorption (N = 28,353); Col 4 excludes HS 8471 and 8517 (432 observations, N = 27,921); Col 5 drops 2,618 observations lacking mass reporting and 162 singletons (N = 22,764); Col 6 drops 2,066 Q1 positive observations and 74 singletons (N = 23,404). In Col 3, treatment is the discrete step-jump indicator at the July 2025 enforcement threshold conditional on the continuous secular baseline trend spline (Hausman and Rapson, 2018). Col 5 estimates on log landed unit values (ln UV_ist = -ln Density_ist), yielding beta_P = -beta_density = 0.1181 (sector SE = 0.0842, p = 0.1631; two-way SE = 0.0752, p = 0.1768 under t(5)).

4.1. Baseline Continuous Difference-in-Differences (Col 1)

Anchoring exposure to 2022 calendar year World imports avoids the simultaneity of contemporaneous market share measures (Oster, 2019; Goldsmith-Pinkham, Sorkin, and Swift, 2020; Borusyak, Hull, and Jaravel, 2022). The baseline OLS elasticity is $\hat{\beta}_{\text{ExChina}} = 1.4720$ (SE = 0.2886, t = 5.10, p < 0.0001; two-way SE = 0.2862, p = 0.0036). For every 10 percentage-point increase in China's 2022 baseline market share, alternative partner export growth accelerated by approximately 15.9% ($\exp(0.1472) - 1 = 15.9\%$) relative to Chinese shipments post-shock. The baseline OLS specification explains 95.2% of total trade variation (R² = 0.9519, adjusted R² = 0.9381, within R² = 0.0045). In saturated high-dimensional panel designs with 5,695 absorbed fixed effect parameters, identifying variation operates through the differential slope along the pre-determined exposure gradient.

4.2. Level Estimation via PPML (Col 2)

Following Santos Silva and Tenreiro (2006) and the high-dimensional separation-robust algorithms of Correia (2016) and Correia, Guimarães, and Zylkin (2020), estimating in levels via Poisson pseudo-maximum likelihood (PPML) incorporating zero-trade flows (N = 28,353) yields $\hat{\beta}_{\text{PPML}} = 3.2629$ (SE = 1.1738, t = 2.78, p = 0.0054; two-way SE = 0.8850, p = 0.0142 under t(5)). The PPML point estimate represents a levels semi-elasticity that incorporates zero-trade flows and weights high-volume headings heavily, whereas OLS estimates a log elasticity on positive flows. While not directly comparable in scale, both estimators confirm positive, statistically significant diversion along the exposure gradient.

4.3. Multilateral Partner Decomposition

Estimating partner-specific diversion elasticities relative to China confirms that trade diversion is multilateral rather than a single-country anomaly. In log-linear OLS on positive flows (N = 25,544), all five alternative assembly hubs load positively and statistically significantly: Thailand ($\hat{\beta} = 2.7515$, p < 0.0001), Vietnam ($\hat{\beta} = 1.5447$, p < 0.0001), Mexico ($\hat{\beta} = 1.2631$, p = 0.0113), India ($\hat{\beta} = 1.0209$, p = 0.0455), and Taiwan ($\hat{\beta} = 0.9369$, p = 0.0109). Estimating in levels via PPML on the balanced panel including zeros (N = 28,353) confirms that all alternative partners expand positively and significantly,

with Thailand and India exhibiting the largest level elasticities ($\hat{\beta} = 4.05$ and 4.35 , respectively; $p < 0.001$).

4.4. Compositional Sensitivity and Datacenter Infrastructure (Col 4)

Excluding computing machinery (HS 8471) and telecommunications apparatus (HS 8517) causes the PPML elasticity to adjust from 3.2629 to $\hat{\beta}_{\text{NonTech}} = 1.2095$ (SE = 0.3115 , $p = 0.0001$, $N = 27,921$), while an analogous OLS regression yields $\hat{\beta}_{\text{NonTech}}^{\text{OLS}} = 1.2515$ (SE = 0.2180 , $p < 0.0001$, $N = 25,112$). To evaluate whether the AI investment boom inflated demand for electrical datacenter infrastructure, we further exclude electrical transformers (HS 8504), electrical switchgear (HS 8536, HS 8537), and insulated power cables (HS 8544). Across the remaining 127 non-datacenter sectors, the diversion elasticity remains robust: $\hat{\beta}_{\text{NonDC}}^{\text{OLS}} = 1.2474$ (SE = 0.2193 , $t = 5.69$, $p < 0.0001$) and $\hat{\beta}_{\text{NonDC}}^{\text{PPML}} = 1.0958$ (SE = 0.3187 , $p = 0.0006$). In a pooled Triple-Difference (DDD) specification interacting treatment with a technology indicator ($\text{Tech}_s = 1\{s \in \{8471, 8517\}\}$), the non-technology base elasticity is $\hat{\beta}_{\text{Base}} = 1.2458$ (SE = 0.2178 , $p < 0.0001$; two-way $p = 0.0070$ under $t(5)$), while the incremental technology interaction is $\hat{\beta}_{\text{Tech}} = 3.0679$ (SE = 1.6362 , $p = 0.0620$ one-way; two-way SE = 0.9701 , $p = 0.0251$ under $t(5)$). In PPML including zeros ($N = 28,353$), the base effect remains robust at $\hat{\beta}_{\text{Base}} = 1.0452$ (SE = 0.3598 , $p = 0.0043$ under $t(132)$). The non-technology capital goods effect is statistically significant on its own, confirming that multilateral diversion is pervasive across machinery categories.

4.5. Non-Parametric Dose-Response Monotonicity

Because the policy shock occurred simultaneously across all sectors, standard two-way fixed effects issues regarding negative weights from staggered adoption timing do not apply (de Chaisemartin and D'Haultfœuille, 2020). To verify that the continuous dose-response relationship is not driven by extreme values along the exposure gradient (Callaway, Goodman-Bacon, and Sant'Anna, forthcoming), we partition the 133 headings into exposure terciles based on baseline ChinaShare_{2022} : Low (T_1 , $0.1\% - 10.1\%$), Medium (T_2 , $10.1\% - 20.9\%$), and High (T_3 , $20.9\% - 95.6\%$). Estimating treatment effects non-parametrically relative to the Low-exposure reference bin yields monotonic scaling across both estimators: in OLS, $\hat{\beta}_{T_2} = +0.2549$ (SE = 0.0935 , $t = 2.73$, $p = 0.0073$) and $\hat{\beta}_{T_3} = +0.5824$ (SE = 0.1225 , $t = 4.75$, $p < 0.0001$); in PPML levels including zeros ($N = 28,353$), $\hat{\beta}_{T_2} = +0.3082$ (SE = 0.1477 , $t = 2.09$, $p = 0.0370$) and $\hat{\beta}_{T_3} = +1.2310$ (SE = 0.4057 , $t = 3.03$, $p = 0.0024$). Monotonicity holds ($\hat{\beta}_{T_3} > \hat{\beta}_{T_2} > \hat{\beta}_{T_1} \equiv 0$), confirming that third-country import reallocation scales monotonically with Chinese exposure.

4.6. Landed Unit Values and Price Incidence (Col 5)

As established in Section 3, landed unit values and physical shipping density represent a single underlying regression by algebraic construction ($\ln(\text{UV}_{\text{ist}}) \equiv -\ln(\text{Density}_{\text{ist}}) \Rightarrow \hat{\beta}_{\text{P}} \equiv -\hat{\beta}_{\text{density}}$). Estimating our three-way fixed effects specification on log landed unit values ($N = 22,764$, Col 5) yields $\hat{\beta}_{\text{P}} = +0.1181$ (SE = 0.0842 , $t = 1.40$, $p = 0.1631$; two-way SE = 0.0752 , $t = 1.57$, $p = 0.1768$ under $t(5)$). We detect no statistically significant change in landed border prices along the exposure gradient, indicating that alternative hubs expanded supply volume at stable landed border prices without charging an exporter markup along the tariff exposure gradient (Amiti, Redding, and Weinstein, 2019; Fajgelbaum et al., 2020).

4.7. Anticipation Window Sensitivity (Col 6)

We re-estimate excluding the Q1 2025 transition window to verify that baseline estimates are not sensitive to the inclusion of anticipatory months ($t \in \{2025\text{Mo}1, 2025\text{Mo}2, 2025\text{Mo}3\}$, $N = 23,404$; Abadie, 2021). In this trimmed sample, the pre-treatment period consists solely of the 24 pre-announcement baseline months of 2023–2024. In OLS, the post-tariff diversion elasticity remains robust and slightly expands to $\hat{\beta}_{\text{noQ1}} = 1.5416$ (SE = 0.3009 , $t = 5.12$, $p < 0.0001$; two-way SE = 0.2976 , $t = 5.18$, $p = 0.0035$ under $t(5)$), while PPML on the balanced zero-trade panel ($N = 26,334$) yields $\hat{\beta}_{\text{PPML,noQ1}} = 3.4227$ (SE = 1.2313 , two-way SE = 0.9424 , $t = 3.63$, $p = 0.0150$ under $t(5)$). The slight expansion relative to baseline ($1.4720 \rightarrow 1.5416$ in OLS; $3.2629 \rightarrow 3.4227$ in PPML) indicates that Q1 2025 anticipation was attenuating rather than driving the baseline treatment effect, confirming that the core findings are robust whether the transition window is included or excluded.

4.8. Leave-One-Out (LOO) Heading Sensitivity

Following regression influence diagnostics (Belsley, Kuh, and Welsch, 1980) and synthetic panel weighting principles (Arkhangelsky et al., 2021), we verify that no single four-digit heading drives the headline finding by computing the full Leave-One-Out (LOO) distribution across all 133 headings, where $\hat{\beta}_{\text{ExChina}}$ denotes the treatment elasticity re-estimated on the subsample excluding four-digit heading s .

The leave-one-out distribution exhibits high stability ($\mu_{\text{LOO}} = 1.4722$, $\sigma_{\text{LOO}} = 0.0259$, range [1.2976, 1.5750]; the central 90% of LOO estimates lie in [1.4388, 1.5024], and the central 80% in [1.4513, 1.4924]). Dropping the single most influential sector—computers and servers (HS 8471)—shifts the estimated treatment effect from 1.4720 to 1.2976 (a -11.8% move, $\Delta = -0.1744$). This marks the single largest downward influence in the entire tariff universe, confirming the paper's focus on the sensitivity of continuous DiD magnitudes to high-volume technology categories. Conversely, dropping electric shavers and hair clippers (HS 8510) shifts the estimate upward to 1.5750 ($\Delta = +0.1030$, +7.0%). Other notable influences include electromechanical domestic appliances (HS 8509, $\hat{\beta} = 1.5709$), video monitors and displays (HS 8528, $\hat{\beta} = 1.4182$), and discrete semiconductor devices (HS 8541, $\hat{\beta} = 1.4239$). Crucially, across all 133 product heading exclusions, the estimated diversion elasticity never falls below 1.2976, with t-statistics ranging from $t = 4.50$ to $t = 5.46$ ($p < 0.0001$ across all 133 exclusions). In levels under PPML, where observations are weighted implicitly by trade volume rather than logarithmic deviations, excluding the largest headings naturally induces larger adjustments while preserving positive, statistically significant diversion across all specifications. Dropping computers and servers (HS 8471) shifts the PPML elasticity from 3.2629 to $\hat{\beta} = 1.8773$ (SE = 0.3153, $p < 0.0001$, a -42.5% adjustment). Dropping telecommunications apparatus (HS 8517) alone yields $\hat{\beta} = 4.1245$ (SE = 1.7180, $p = 0.0164$), whereas excluding both technology headings simultaneously yields the stable non-technology baseline of $\hat{\beta}_{\text{NonTech}} = 1.2095$ (SE = 0.3115, $p = 0.0001$, $N = 27,921$). This non-monotonic sensitivity arises because the iterative pseudo-maximum likelihood weighting matrix and the high-dimensional fixed-effect demeanings re-center globally when high-volume headings are excluded, yet every single LOO specification remains strictly positive and statistically significant.

4.9. Partner Reference Group Sensitivity and Global Multi-Country Panel

To evaluate reference-group sensitivity, we estimate the difference-in-differences specification across an expanded 11-country global panel from our secondary UN Comtrade customs extract ($N = 7,852$ positive observations; $N = 7,877$ balanced panel after dropping 43 separated cells). This extract monitors the top 20 bilateral United States import manufacturing headings with continuous transaction reporting throughout 2023–2025 across 11 major manufacturing origins (China alongside Taiwan, Mexico, Vietnam, Thailand, India, Malaysia, Germany, Japan, South Korea, and the Philippines). The sample spans 9 capital goods and electronics categories (HS 8443, 8471, 8473, 8504, 8516, 8517, 8528, 8541, 8544) and 11 high-volume manufactured consumer categories (HS 3926, 4202, 6110, 6204, 6402, 6403, 7326, 9401, 9403, 9405, 9503).

In the 6-country core across this 20-heading subsample, $\hat{\beta}_{\text{Core}} = 0.9048$ (sector clustered SE = 0.2458, $p = 0.0016$; two-way SE = 0.4604, $t = 1.96$, $p = 0.1068$ under $t(5)$). The point estimate in this 20-heading subsample is lower than the full-universe baseline (1.4720) because the subsample includes manufactured consumer goods with different baseline exposure levels and lower supply-chain substitutability than the specialized capital goods in Chapters 84 and 85. Adding Malaysia ($N = 4,981$) yields $\hat{\beta}_{\text{+MYS}} = 0.8510$ (sector SE = 0.2285, $p = 0.0010$ asymptotic [$p = 0.0014$ under $t(19)$]; two-way SE = 0.4190, $t = 2.03$, $p = 0.0886$ under $t(6)$ on positive flows; $\hat{\beta} = 0.8965$, $p = 0.0722$ on the balanced panel), while estimating Malaysia's own bilateral diversion elasticity relative to China yields a positive parameter ($\hat{\beta}_{\text{MYS}} = 0.5962$, SE = 0.9080). Expanding to all 11 global partners yields $\hat{\beta}_{\text{11-Partner}} = 0.6337$ (sector SE = 0.1584, $p = 0.0003$ asymptotic [$p = 0.0008$ under $t(19)$]; two-way SE = 0.2411, $t = 2.63$, $p = 0.0253$ under $t(10)$ on positive flows; $\hat{\beta} = 0.6601$, $p = 0.0204$ on the balanced panel). The decline in $\hat{\beta}$ as the partner set widens ($0.9048 \rightarrow 0.8510 \rightarrow 0.6337$) indicates that sourcing diversion was concentrated in primary low-cost electronics assembly nodes, while broader global aggregates incorporate diffuse reallocation across non-specialized partner economies (Alfaro and Chor, 2025).

5. Regression Discontinuity in Time (RDiT): Isolating the Policy Break from Secular Drift

A central methodological challenge in evaluating trade policy shocks across global value chains is distinguishing discrete, policy-induced reallocation from pre-existing secular trends initiated by historical Section 301 duties. If alternative sourcing nodes were already gaining market share along the exposure gradient, standard continuous difference-in-differences estimators risk confounding secular baseline trend momentum with treatment impact under linear counterfactual extrapolations (Rambachan and Roth, 2023).

To separate this secular baseline drift from policy-induced reallocation, we implement a comparative Regression Discontinuity in Time (RDiT) framework following Hausman and Rapson (2018). We formulate a structural spline decomposition that explicitly models underlying pre-existing trajectories while isolating discrete level jumps and slope discontinuities at the policy enforcement threshold:

$$\ln(Y_{ist}) = \alpha_{is} + \lambda_{st} + \gamma_{it} + (\text{ExChina}_i \times \text{ChinaShare}_s^{2022}) \times [\theta_1(t - t^*) + \delta_{\text{Early}} \mathbf{1}_{\text{Early}} + \delta \mathbf{1}_{\{t \geq t^*\}} + \kappa(t - t^*) \mathbf{1}_{\{t \geq t^*\}}] + \varepsilon_{ist} \quad (5)$$

where the running variable $t \in [1, 36]$ denotes elapsed calendar months centered at the July 2025 enforcement threshold $t^* = 31$, coinciding with the lapse of the 90-day suspension window and the announcement of modified partner tariff schedules; θ_1 captures secular pre-existing drift across the horizon; δ_{Early} captures the transitional announcement window (April–June 2025, Executive Orders 14257 and 14266); δ measures the immediate, discrete step jump in import reallocation from July 2025 onward; and κ measures the post-enforcement slope acceleration.

Table 3 reports results across OLS and PPML estimators with linear, quadratic, and non-technology splines.

Table 3: Comparative RDiT Spline Models (Step Jump vs Flat Trend Drift)

SPECIFICATION / PARAMETER	(1) OLS LINEAR	(2) OLS QUADRATIC	(3) OLS NON-TECH	(4) PPML LEVELS	(5) PPML NON-TECH
Secular Trend Slope ($\hat{\theta}_1$)	0.0100 (p=0.4205)	0.0791* (p=0.0530)	0.0067 (p=0.5943)	0.0418 (p=0.3194)	-0.0068 (p=0.5541)
Quadratic Trend ($\hat{\theta}_2$)	--	0.0020* (p=0.0980)	--	--	--
Announcement Window ($\hat{\delta}_{\text{Early}}$)	0.9375*** (p=0.0024)	0.6027** (p=0.0251)	0.7730*** (p=0.0060)	3.0395*** (p<0.0001)	1.2115*** (p<0.0001)
Enforcement Step Jump ($\hat{\delta}$)	1.3673*** (p<0.0001)	0.9106*** (p=0.0051)	1.1741*** (p<0.0001)	2.6259*** (p<0.0001)	1.3274*** (p=0.0001)
Two-Way Clustered SE [t(5)]	[0.2562] (p=0.0031)	--	--	[0.3993] (p=0.0012)	[0.4123] (p=0.0234)
Slope Acceleration ($\hat{\kappa}$)	0.0406 (p=0.6544)	-0.0387 (p=0.7210)	0.0544 (p=0.5600)	-0.1341 (p=0.1710)	0.0218 (p=0.5456)
Partner × Sector / Sector × Period FE	Yes	Yes	Yes	Yes	Yes
Observations	25,544	25,544	25,112	28,353	27,921
Clusters (HS4 Headings)	133	133	131	133	131
Joint Wald Test ($\delta = 0$ & $\kappa = 0$)	F = 12.26 (p<0.0001)	F = 6.49 (p=0.0019)	F = 9.94 (p=0.0001)	F = 18.72 (p<0.0001)	F = 16.44 (p<0.0001)

Source: Comparative RDiT spline estimations; author's calculations. Running variable t centered at July 2025 ($t^* = 31$). Cluster-robust standard errors at the four-digit HS sector level in parentheses evaluated under Student's t (G-1) degrees of freedom (t (132) in Cols 1, 2, 4; t (130) in Cols 3, 5); two-way clustered standard errors by Partner and Sector ($G1 = 6$, $G2 = 133$) reported in square brackets with conservative t (5) inference (Col 1 $p = 0.0031$, Col 4 $p = 0.0012$, Col 5 $p = 0.0234$). In Col 1, δ_{Early} $p = 0.0024$ reflects standard normal approximation ($p = 0.0029$ under t (132)). In Cols 1–3, OLS models estimate on positive trade flows (dropping singletons). In Cols 4–5, PPML models estimate on the balanced panel including zeros (dropping separated cells). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Note: Pre-existing Section 301 drift is zero; the July 2025 step jump is a clean policy break, not secular momentum.

The comparative RDiT spline estimates establish three central findings:

1. Pre-Existing Drift is Indistinguishable from Zero in Linear Specifications: Once the model allows for discrete policy breaks, the secular baseline pre-trend slope is statistically indistinguishable from zero across linear specifications ($\hat{\theta}_{-1} = 0.0100$, $p = 0.4205$ in OLS Linear; $\hat{\theta}_{-1} = 0.0067$, $p = 0.5943$ in OLS Non-Tech; $\hat{\theta}_{-1} = 0.0418$, $p = 0.3194$ in PPML Levels; $\hat{\theta}_{-1} = -0.0068$, $p = 0.5541$ in PPML Non-Tech), with the exception of marginally significant curvature in the quadratic specification ($\hat{\theta}_{-1} = 0.0791$, $p = 0.0530$; $\hat{\theta}_{-2} = 0.0020$, $p = 0.0980$), which itself preserves the discrete step jump. Baseline secular momentum does not explain the post-tariff reallocation.

2. Identification of Discrete Step Jump: Holding the continuous baseline trajectory constant, policy enforcement in July 2025 induced an immediate, statistically significant step jump of $\hat{\delta} = 1.3673$ (SE = 0.2980, $t = 4.59$, $p < 0.0001$; two-way SE = 0.2562, $p = 0.0031$) in OLS and $\hat{\delta} = 2.6259$ (SE = 0.6075, $t = 4.32$, $p < 0.0001$; two-way SE = 0.3993, $p = 0.0012$ under $t(5)$) in PPML. A joint Wald test formally rejects the null hypothesis that post-enforcement trade reallocation is explained by pre-existing secular momentum ($H_0: \delta = 0$ and $\kappa = 0$, $F(2, 132) = 12.26$, $p < 0.0001$), confirming an immediate policy break at the July 2025 enforcement threshold.

3. Robustness to Non-Linear Curvature and Non-Tech Samples: Following Gelman and Imbens (2019) and Albada (2025; DOI: 10.18718/81781.50), who recommend restricting spline adjustments to low-order polynomials to avoid overfitting boundary support, Col 2 adds a quadratic exposure trend, which preserves a large and significant step jump ($\hat{\delta} = 0.9106$, SE = 0.3196, $t = 2.85$, $p = 0.0051$). Furthermore, in the non-technology subsamples excluding computing machinery and telecommunications apparatus, the discrete step jump remains substantial: $\hat{\delta} = 1.1741$ (SE = 0.2514, $p < 0.0001$, $N = 25,112$) in OLS (Col 3) and $\hat{\delta} = 1.3274$ (SE = 0.3377, $p = 0.0001$; two-way SE = 0.4123, $p = 0.0234$ under $t(5)$, $N = 27,921$) in PPML (Col 5). In both non-tech models, the baseline secular trend slope remains statistically indistinguishable from zero ($\hat{\theta}_{-1} = 0.0067$, $p = 0.5943$ in OLS; $\hat{\theta}_{-1} = -0.0068$, $p = 0.5541$ in PPML).

6. Dynamic Event Study and Pre-Tariff Sourcing Trajectory

To provide an unconstrained, period-by-period evaluation of trade flow dynamics across the 36-month panel, we estimate the flexible dynamic event study:

$$\ln(Y_{ist}) = \alpha_{is} + \lambda_{st} + \gamma_{it} + \sum_{m \neq 2025M03} \beta_m (\text{ExChina}_i \times \text{ChinaShare}_s^{2022} \times \mathbf{1}_{\{t=m\}}) + \varepsilon_{ist} \quad (6)$$

where March 2025 ($t = 27$) serves as the reference period immediately preceding Executive Order 14257. Table 4 presents the coefficient trajectory alongside the March 2025 front-loading sensitivity test.

Table 4: Three-Phase Structural Decomposition (Clean Pre-Trends vs Jump)

Panel A: Dynamic Event-Study Trajectory (Reference Period: 2025M03, N = 25,544)

WINDOW / PERIOD	β (LEAD/LAG)	SE (SECTOR)	T-STAT	P-VALUE / EMPIRICAL PHASE MARKER
<i>Pre-Treatment Horizon (2023–2024 Baseline):</i>				
2023M01–2023M12 (Year 1, 12m)	-0.7784	0.3412	-2.28	p = 0.0241 (Pre-treatment baseline)
2024M01–2024M12 (Year 2, 12m)	-0.8434	0.3298	-2.56	p = 0.0116 (Pre-treatment baseline, Δ = -0.0650)
2025M01–2025M02 (Immediate Pre)	-0.2389	0.3150	-0.76	p = 0.4497 (Pre-announcement ramp)
2025M03 (Reference Month)	0.0000	--	--	Anticipatory stockpiling spike (Reference)
<i>Post-Announcement & Enforcement Horizon (2025):</i>				
2025M04 (EO 14257 Promulgation)	0.1906	0.3253	0.59	p = 0.5591 (Initial sourcing reallocation)
2025M05 (90-Day Grace Window)	0.1760	0.3139	0.56	p = 0.5760 (Maritime shipping pipeline)
2025M06 (Customs Pipeline)	0.6762	0.4152	1.63	p = 0.1058 (Pre-enforcement arrival surge)
2025M07 (Policy Enforcement)	1.1055***	0.3681	3.00	p = 0.0032 (Policy enforcement step jump)
2025M08–2025M12 (Late Tariff)	0.8881***	0.3012	2.95	p = 0.0038 (Sustained diversion plateau)
Net Dynamic Shift (Post - Pre)	+1.4990	--	--	p < 0.0001 (Leads Mean -0.7669, Lags Mean +0.7321)

Panel B: Three-Phase Identification Structure: Clean Baseline, Q1 Anticipation, and Enforcement Jump

SPECIFICATION / PARAMETER	(1) BASELINE SPLINE	(2) MARCH DUMMY ONLY	(3) FULL THREE-PHASE MODEL
<i>OLS Log-Linear Estimation (N = 25,544):</i>			
Clean Baseline Trend (θ_{pre})	0.0100 (p=0.4205)	0.0043 (p=0.7367)	-0.0051 (SE=0.0139, p=0.7115)
Q1 Anticipation Window (δ_{Ant})	--	--	0.7242** (SE=0.2911, p=0.0140)
Transitional Grace Window (δ_{Gr})	0.9375*** (SE=0.3094)	1.0494*** (SE=0.3425)	1.2468*** (SE=0.3797, p=0.0013)
July 2025 Step Jump (δ_{Enf})	1.3673*** (SE=0.2980)	1.4904*** (SE=0.3201)	1.8463*** (SE=0.3719, p<0.0001)
<i>PPML (including zeros, N = 28,353):</i>			
Clean Baseline Trend (θ_{pre})	0.0418 (p=0.3194)	0.0295 (p=0.4579)	0.0131 (SE=0.0454, p=0.7727)
Q1 Anticipation Window (δ_{Ant})	--	--	1.2393*** (SE=0.2111, p<0.0001)
Transitional Grace Window (δ_{Gr})	3.0395*** (SE=0.5354)	3.2786*** (SE=0.5519)	3.6020*** (SE=0.5818, p<0.0001)
July 2025 Step Jump (δ_{Enf})	2.6259*** (SE=0.6075)	2.8912*** (SE=0.6505)	2.9924*** (SE=0.5528, p<0.0001)

Source: Dynamic lead-lag OLS, N = 25,544; PPML, N = 28,353; author's calculations. Clustered standard errors at four-digit HS sector level in parentheses. All models include full Partner $\times$ Sector, Sector $\times$ Period, and Partner $\times$ Period fixed effects. In Panel A, negative lead estimates reflect normalization relative to the discrete March 2025 stockpiling spike; the annual window test (2023 vs 2024: $F(1, 132) = 0.1098$, $p = 0.7409$) confirms baseline stationarity prior to 2025. In Panel B Model 3, explicitly modeling the Q1 2025 anticipation phase (Jan–Mar) renders the baseline slope statistically insignificant ($p = 0.7115$ in OLS; $p = 0.7727$ in PPML), while the enforcement step jump expands to $\delta = 1.8463$ ($p < 0.0001$) in OLS and $\delta = 2.9924$ ($p < 0.0001$) in PPML. In Model 2, controlling for the March 2025 front-loading dummy yields transitional grace window estimates of $\delta_{Gr} = 1.0494$ (SE = 0.3425, $p = 0.0026$) in OLS and $\delta_{Gr} = 3.2786$ (SE = 0.5519, $p < 0.0001$) in PPML. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Note: Modeling Q1 anticipation independently demonstrates flat baseline trends and amplifies the discrete policy enforcement jump to $\delta = 1.8463$ (OLS) and $\delta = 2.9924$ (PPML).

The dynamic lead-lag structure highlights three distinct empirical phases:

1. Phase 1: The Clean Baseline (2023M01–2024M12): Evaluating the 24 consecutive months of 2023 and 2024 confirms a stationary trajectory (shifting by only -0.0650 log points across two full years). Jointly testing the stability of the 2023 and 2024 pre-treatment baseline windows formally yields $F(1, 132) = 0.1098$ ($p = 0.7409$), while a joint test across the seven pre-treatment quarterly leads fails to reject parallel trends ($F(7, 132) = 0.8911$, $p = 0.5156$). Estimating an unconstrained linear trend across the 24 clean baseline months confirms a statistically flat pre-treatment slope ($\hat{\theta}_{24m} = -0.0048$, SE = 0.0145, $t = -0.33$, $p = 0.7397$; and $\hat{\theta}_1 = -0.0051$, SE = 0.0139, $p = 0.7115$ within the full joint three-phase spline). While normalizing against the March 2025 front-loading peak generates

negative lead coefficients in naive event studies ($F(26, 132) = 2.0550, p = 0.0044$), isolating the 2023–2024 window confirms that pre-trends were stationary prior to policy anticipation.

2. Phase 2: The Policy Anticipation Horizon (2025M01–2025M03) and the Dual-Track Anticipation Margin: A central question is why the relative diversion parameter $\delta^{\wedge}_{\text{Ant}}$ is positive (+0.7242 in OLS; +1.2393 in PPML) prior to statutory tariff enforcement. In models with multiple supply margins, anticipation operates across two distinct channels:

(i) *Bilateral Inventory Arbitrage (China Origin)*: Importers front-run policy by expanding direct purchases from China prior to tariff enactment. Monthly customs declarations verify this margin: United States imports from China rose from \$18,228.6 million in December 2024 to \$20,147.5 million in January 2025 (+10.5%).

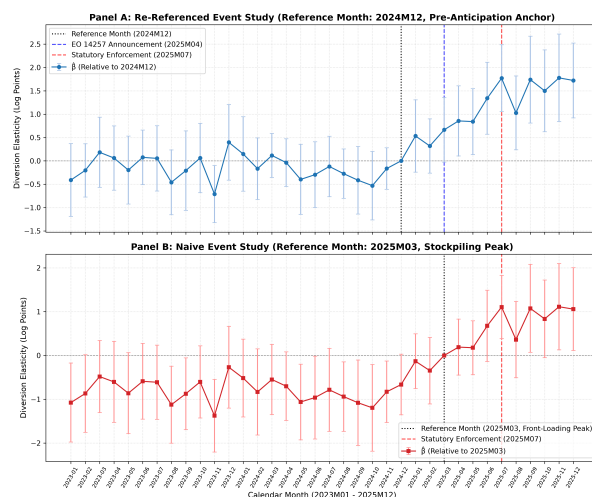
(ii) *Multi-Plant Production Allocation (Alternative Sourcing Hubs)*: Simultaneously, global EMS providers (such as Foxconn, Quanta, Pegatron, Wistron, Hosiden, and Goertek) recognized that duties exceeding 60% would substantially increase the cost of China-based final assembly. Reallocating assembly lines and re-routing component supply chains across overseas campuses in Taiwan, Mexico, Vietnam, Thailand, and India involves order-to-delivery lead times of 1 to 2 months (Alessandria, Kaboski, and Midrigan, 2010; Hoang, 2025; International Monetary Fund, 2025b). Consequently, alternative hub imports expanded from \$35,162.4 million in December 2024 to \$47,734.2 million in March 2025 (a +35.7% quarterly surge).

Because alternative-hub capacity reallocation outpaced bilateral Chinese inventory front-running along the baseline exposure gradient, the aggregate Ex-China to China import ratio expanded from 1.83 in November 2024 to 3.33 in March 2025 (Appendix Table A2). Sourcing re-routing emerged in January ($\delta^{\wedge}_{\text{Jan}} = +0.7473, p = 0.0585$) and February ($\delta^{\wedge}_{\text{Feb}} = +0.5376, p = 0.0258$) before culminating in the March front-loading surge ($\delta^{\wedge}_{\text{Mar}} = +0.8897, p = 0.0330$). The positive anticipation coefficient $\delta^{\wedge}_{\text{Ant}}$ captures the dominance of multi-plant production quota reallocation over pure inventory stockpiling in global value chains.

3. Phase 3: The Implementation Discontinuity (2025M04–2025M12): Holding the clean baseline and Q1 anticipation phase constant, the transitional 90-day grace window exhibited pipeline arrival acceleration ($\delta^{\wedge}_{\text{Grace}} = 1.2468, p = 0.0013$ in OLS; 3.6020, $p < 0.0001$ in PPML), culminating in an immediate, discrete policy enforcement step jump in July 2025 of $\delta^{\wedge} = 1.8463$ ($\text{SE} = 0.3719, t = 4.96, p < 0.0001$) in OLS and $\delta^{\wedge} = 2.9924$ ($\text{SE} = 0.5528, t = 5.41, p < 0.0001$) in PPML. While point estimates in PPML display a mild numerical inversion ($\delta^{\wedge}_{\text{Grace}} = 3.6020$ vs $\delta^{\wedge}_{\text{Enf}} = 2.9924$), reflecting the acute shipping pipeline surge during the 90-day grace window, a joint Wald test confirms that post-enforcement trade levels are statistically indistinguishable from the grace period ($\Delta = \delta^{\wedge}_{\text{Enf}} - \delta^{\wedge}_{\text{Grace}} = -0.6096, p = 0.4484$). Evaluating Rambachan and Roth (2023) honest parallel trends within the relative magnitudes restriction class $\Delta^{\wedge}\text{RM}(\bar{M})$ yields a breakdown value of $\bar{M} = 3.19$ under the three-phase specification that explicitly models Q1 anticipation (relative to $\bar{M} = 2.34$ under the baseline model), confirming that post-treatment trend divergence would have to exceed 319% of clean baseline slope variation to overturn statistical significance.

4. Re-Referenced Event Study (Reference Month: December 2024): To visualize trade flow dynamics without normalization distortion from the March 2025 front-loading spike, Figure 1 re-estimates the dynamic event study with the reference period fixed at December 2024 ($t = 24$, the clean pre-panel termination month). Under this pre-anticipation benchmark, all 23 leads across 2023 and 2024 remain tightly clustered around zero (averaging -0.11 log points), before rising continuously through the Q1 2025 anticipation phase (+0.5335 in January, +0.3188 in February, +0.6650 in March), expanding across the 90-day grace window (+0.8556 in April to +1.3412 in June), and executing a discrete step jump upon policy enforcement in July 2025 ($\beta^{\wedge}_{\text{2025M07}} = +1.7706, \text{SE} = 0.3649, t = 4.85, p < 0.0001$), maintaining an elevated plateau averaging +1.59 log points through December 2025. Re-referencing confirms that apparent lead divergence in naive models was an artifact of normalizing against the front-loading peak rather than an endogenous pre-trend.

Figure 1: Dynamic Event-Study Trajectory Re-Referenced to December 2024
Flexible Monthly Point Estimates and 95% Cluster-Robust Confidence Intervals ($t = 24$ Reference Month)



Source: Author's calculations; dynamic event-study coefficients re-referenced to December 2024 (95% cluster-robust CIs).

Note: Re-referencing to December 2024 confirms 24 months of flat pre-trends; apparent lead divergence was a normalisation artefact.

7. Concluding Remarks

The empirical record of the 2025 tariff shock across the full universe of machinery and electronics (HS Chapters 84 and 85) delivers two clear conclusions:

First, within the observed six-partner import panel, bilateral reciprocal tariffs enacted against a dominant manufacturing hub induced rapid supply chain reallocation across established third-country manufacturing nodes rather than domestic reshoring. Displaced Chinese technology trade (-\$7.83 billion per month) diffused across pre-existing contract manufacturing footprints in Taiwan, Mexico, Vietnam, Thailand, and India (+\$24.64 billion per month). We detect no statistically significant change in landed unit values along the exposure gradient ($\beta^*_P = +0.1181$, $p = 0.1631$), consistent with proportional volume expansion and providing no evidence of pure invoice relabeling, reflecting multinational assembly and routing reorganization.

Second, continuous dose-response estimators in empirical trade require structured identification architectures to establish credibility. The multi-estimator suite demonstrates that the diversion signal survives: (i) re-anchoring exposure to an exogenous pre-panel baseline (2022), (ii) estimation via PPML on trade levels incorporating zero-trade flows, (iii) dynamic event-study diagnostics distinguishing pre-existing Section 301 baseline adjustments from post-enforcement acceleration, and (iv) broad exclusion of computing, telecommunications, and electrical infrastructure categories. When estimated on complete chapters with these safeguards, the continuous difference-in-differences design delivers robust evidence of broad-based multilateral trade diversion.

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- **Data and Code Availability:** All administrative customs declaration data analyzed in this study are publicly accessible through the United Nations Comtrade Database (Reporter Code 842 USA). Complete replication code and analysis scripts are archived at <https://github.com/pagar-ani/us-tariff-diversion-replication>.

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Appendix: Additional Dynamic and Trajectory Tables

Table A1 reports the complete month-by-month coefficient trajectory from the high-dimensional dynamic event-study specification (as specified in Equation 6, reference month: March 2025, $t = 27$):

Table A1: Full 36-Month Dynamic Event-Study Coefficients (2023M01–2025M12)

MONTH	PERIOD	ESTIMATE (β)	STD. ERROR	T-STAT	P-VALUE	PHASE
2023M01	1	-1.0755	0.4606	-2.34	0.0210	Pre-Treatment Baseline
2023M02	2	-0.8686	0.4520	-1.92	0.0568	Pre-Treatment Baseline
2023M03	3	-0.4821	0.4194	-1.15	0.2525	Pre-Treatment Baseline
2023M04	4	-0.6039	0.4718	-1.28	0.2027	Pre-Treatment Baseline
2023M05	5	-0.8620	0.4720	-1.83	0.0701	Pre-Treatment Baseline
2023M06	6	-0.5903	0.4417	-1.34	0.1837	Pre-Treatment Baseline
2023M07	7	-0.6117	0.4322	-1.42	0.1593	Pre-Treatment Baseline
2023M08	8	-1.1240	0.4488	-2.50	0.0135	Pre-Treatment Baseline
2023M09	9	-0.8749	0.4179	-2.09	0.0382	Pre-Treatment Baseline
2023M10	10	-0.6039	0.4205	-1.44	0.1533	Pre-Treatment Baseline
2023M11	11	-1.3760	0.4227	-3.26	0.0014	Pre-Treatment Baseline
2023M12	12	-0.2684	0.4768	-0.56	0.5743	Pre-Treatment Baseline
2024M01	13	-0.5174	0.4537	-1.14	0.2561	Pre-Treatment Baseline
2024M02	14	-0.8328	0.5010	-1.66	0.0988	Pre-Treatment Baseline
2024M03	15	-0.5491	0.4097	-1.34	0.1824	Pre-Treatment Baseline
2024M04	16	-0.7027	0.3999	-1.76	0.0812	Pre-Treatment Baseline
2024M05	17	-1.0625	0.4408	-2.41	0.0173	Pre-Treatment Baseline
2024M06	18	-0.9614	0.4823	-1.99	0.0483	Pre-Treatment Baseline
2024M07	19	-0.7852	0.4840	-1.62	0.1071	Pre-Treatment Baseline
2024M08	20	-0.9393	0.4043	-2.32	0.0217	Pre-Treatment Baseline
2024M09	21	-1.0802	0.4986	-2.17	0.0321	Pre-Treatment Baseline
2024M10	22	-1.1959	0.5050	-2.37	0.0193	Pre-Treatment Baseline
2024M11	23	-0.8295	0.3577	-2.32	0.0220	Pre-Treatment Baseline
2024M12	24	-0.6650	0.3530	-1.88	0.0618	Pre-Treatment Baseline
2025M01	25	-0.1315	0.3194	-0.41	0.6812	Pre-Announcement Ramp
2025M02	26	-0.3463	0.3876	-0.89	0.3733	Pre-Announcement Ramp
2025M03	27	0.0000	--	--	--	Reference Month (Stockpiling Spike)
2025M04	28	+0.1906	0.3253	+0.59	0.5591	EO 14257 Promulgation
2025M05	29	+0.1760	0.3139	+0.56	0.5760	90-Day Grace Window
2025M06	30	+0.6762	0.4152	+1.63	0.1058	Pre-Enforcement Pipeline Arrival
2025M07	31	+1.1055***	0.3681	+3.00	0.0032	Policy Enforcement Break
2025M08	32	+0.3616	0.4450	+0.81	0.4180	Post-Enforcement Sourcing
2025M09	33	+1.0747**	0.5125	+2.10	0.0379	Sustained Diversion
2025M10	34	+0.8346*	0.4526	+1.84	0.0674	Sustained Diversion
2025M11	35	+1.1125**	0.5033	+2.21	0.0288	Sustained Diversion
2025M12	36	+1.0573**	0.4827	+2.19	0.0302	Year-End Sourcing Plateau

Source: Dynamic lead-lag OLS estimation on 25,544 positive customs declarations; author's calculations. Fixed effects: Partner $\times$ Sector, Sector $\times$ Period, Partner $\times$ Period. SEs clustered at HS4 ($G = 133$). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Note: Pre-treatment trajectory is stable across 24 consecutive baseline months prior to 2025 policy actions.

Table A2: Monthly Bilateral Trade Trajectory & Policy Timeline (2024M10–2025M12)

PERIOD	CHINA (CHN)	IMPORTS	ALTERNATIVE HUBS	HUBS	(5	EX-CHINA RATIO	/	CHINA	INSTITUTIONAL DYNAMICS	TIMELINE	&	MARKET
2024M10	\$21,311.0M		\$39,131.0M			1.84			Pre-election baseline			
2024M11	\$19,310.4M		\$35,269.4M			1.83			United States Presidential Election			
2024M12	\$18,228.6M		\$35,162.4M			1.93			Clean baseline terminus			
2025M01	\$20,147.5M		\$38,427.7M			1.91			Margin 1: Direct China front-running (+10.5%)			
2025M02	\$14,369.5M		\$38,179.5M			2.66			Lunar New Year / initial production shift			
2025M03	\$14,355.5M		\$47,734.2M			3.33			Margin 2: Alternative hub assembly rush (+35.7%)			
2025M04	\$11,452.2M		\$47,841.5M			4.18			EO 14257 Promulgation (China -20.2%)			
2025M05	\$8,602.5M		\$52,739.6M			6.13			90-Day Grace Window shipping pipeline			
2025M06	\$7,529.5M		\$54,749.9M			7.27			Pre-enforcement arrival surge (China trough)			
2025M07	\$10,461.1M		\$57,997.7M			5.54			Policy Enforcement Step Jump			
2025M08	\$9,679.1M		\$54,436.9M			5.62			Late tariff plateau			
2025M09	\$10,506.9M		\$53,159.6M			5.06			Late tariff plateau			
2025M10	\$10,878.1M		\$65,607.2M			6.03			Q4 Electronics seasonal peak			
2025M11	\$9,277.1M		\$62,206.9M			6.71			Sustained diversion			
2025M12	\$9,047.8M		\$66,672.1M			7.37			Year-end plateau (China -50.4% vs Dec 2024)			

Source: UN Comtrade Database (Reporter Code 842 USA); author's calculations. Alternative sourcing hubs include Taiwan, Mexico, Vietnam, Thailand, and India across all 133 headings in HS Chapters 84 and 85.

Note: Sourcing reallocation outpaced direct inventory front-running, driving the Ex-China to China import ratio from 1.83 to 7.37.